Abstract

Manual synthesis cannot keep pace with a fast-growing research literature, and ad-hoc reviews bind no evidence to a reproducible pipeline. We present a configurable, reproducible meta-analysis framework that takes a single search term and produces a complete quantitative portrait of its literature. For this instance the term is **Modafinil**. The pipeline dispatches across 10 literature engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, ChinaRxiv, Europe PMC, bioRxiv, and medRxiv), each degrading gracefully to a skipped source when an API key or the network is unavailable, then merges and de-duplicates records by a canonical identifier hierarchy (DOI $>$ arXiv ID $>$ Semantic Scholar ID $>$ OpenAlex ID $>$ title digest) into a corpus of $N = 2334$ records spanning 2000–2026 (26 years). Records are classified into a configurable 6-bucket subfield taxonomy (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience); the largest subfield is **Clinical Sleep** (63.0% of the classified corpus). The corpus grows at a compound annual rate of 5.48% (mean year-over-year growth 7.8%, doubling time 9.2 years), peaking in 2025 with 147 records.

Non-negative matrix factorization extracts 5 latent topics over a 500-feature vocabulary, offline deterministic embeddings place every title, abstract, and (when available) full text in a shared vector space, and citation-network analysis exposes the corpus's internal structure (8,623 intra-corpus edges across 2234 nodes, 1416 communities, graph density 0.17%). Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus. Abstract coverage stands at 61.6%, 24.6% of records are open access, and 54.6% have a direct PDF link. An optional, LLM-gated knowledge-graph stage scores the 6 hypotheses explored against the evidence. This run produced 21 publication-quality figures.

Every domain-specific value in this manuscript — the search term, keyword set, engine roster, subfield taxonomy, and hypotheses — is injected from a single configuration file and the pipeline's own outputs; re-targeting the configuration re-targets the entire paper. The result is a reusable architecture for *living literature reviews*: continuously re-runnable, evidence-bound syntheses for any topic.

Keywords: modafinil, meta-analysis, literature retrieval, bibliometrics, record de-duplication, full-text mining, document embeddings, citation network, topic modeling, entity extraction, wakefulness, cognitive enhancement, reproducible research